RFID Reader Communication Protocol

V1.0

1. Protocol Overview

This communication protocol aims to standardize the data interaction process between the Host and the Reader to achieve efficient and accurate control of the Reader, ensuring system stability and reliability.

2. Physical Layer Specification

This communication protocol is applicable to device interfaces including RS232, RS485, USB, RJ45, WiFi, etc.

2.1. RS232

The baud rate selectable range is 9600bps - 921600bps, data bits are 8 bits, stop bit is 1 bit, parity bit selectable range is None, Odd, Even, ensuring consistency in serial port parameters for stable data transmission.

The default factory setting is 115200, N, 8, 1.

2.2. RS485

The baud rate selectable range is 9600bps - 921600bps, data bits are 8 bits, stop bit is 1 bit, parity bit selectable range is None, Odd, Even, ensuring consistency in serial port parameters for stable data transmission.

The default factory setting is 115200, N, 8, 1.

2.3. USB

Supports WinUSB V2.

2.4. RJ45

Supports TCP Server mode, UDP Server mode, TCP Client mode, UDP Client mode communication. The default communication mode is TCP Server mode, server port 8090.

2.5. WiFi

Supports TCP Server mode, TCP Client mode communication. The default communication mode is TCP Server mode, server port 8090.

3. Data Frame Format

3.1. Frame Format

SOF	LEN	ADDR	SEQ	CMD	STATUS	PLYLOAD	CRC
0xFE	2 Bytes	1 Byte	1 Byte	1 Byte	1 Byte	N Bytes	2 Bytes

- Start of Frame (SOF): Fixed 1 byte, value is 0xFE, used by the receiver to quickly identify the start of the data frame and synchronize data reception.
- Total Length (LEN): 2 bytes, value range 0-1024, allows the receiver to precisely control the amount of data received, preventing data overflow or loss; LEN is calculated from SOF to the end of CRC.
- Device Address (ADDR): Fixed 1 byte, 1-254 used to distinguish different devices on the link, address 0xFF is the broadcast address. 0x00 is invalid.
- Sequence Number (SEQ): Fixed 1 byte, sender value range 0-127, used to distinguish different frames; the responder must set this to sender's SEQ + 128; otherwise, it is considered an invalid response.
- Status (STATUS): Fixed 1 byte, fixed to 0 when sending, filled by the responder based on command execution status.
- Command (CMD): 1 byte, specifies the operation instruction for this frame, enabling the device to execute the corresponding function according to the instruction.
- Data Parameters (PAYLOAD): Length depends on the command.
- Frame Check (CRC): Uses CRC-16/CCITT-FALSE algorithm for verification, polynomial 0x1021, initial value 0xFFFF. It verifies from the Start of Frame to the end of the Data Parameters, ensuring data integrity during transmission. The receiver uses this to verify data accuracy and detect transmission errors.

3.2. Data Transmission Format

Data in this protocol is transmitted in Big-endian mode.
The data definitions used in this protocol are as follows:

Data Type	Description
bool	1-byte bool type
byte	1-byte data
short	2-byte data
int	4-byte data
long	8-byte data
float	4-byte single-precision floating point data
byte[N]	byte array
short[N]	short array
int[N]	int array
long[N]	long array
float[N]	float array
array	Variable-length array, LV structure, L occupies 1 byte, describes the array length (in bytes), V is the array content. Maximum array length 255 bytes.
string	Variable-length array, LV structure, L occupies 1 byte, describes the array length (in bytes), V is the array content. Maximum array length 255 bytes.
bitmap[N]	Bitmap definition, occupies (N+7)/8 bytes; bit0 is at byte[0] bit7, bit7 is at byte[0] bit0, and so on.

4. Commands and Responses

4.1. Device Management Commands [0x00~0x5F]

4.1.1. Connection Detection (0x00)

This command is used to detect if the current connection is valid. Can be initiated by either the Reader or the Host.

✓ Request Payload

Parameter	Format	Description
Confirmation Code	int	Confirmation code managed by the initiator

✓ Response Payload

Parameter	Format	Description
Beep Duration	int	Controls the buzzer sounding time, Unit: ms
Parameter	Format	Description
Response Code	int	Bitwise inverse of the Confirmation Code

Frame Example:：

T> FE000D010100000000000104ED
R< FE000D01810000FFFFFFFE36DB

4.1.2. Device Restart（0x01）

This command is used to restart the Reader. The Reader should restart immediately upon receiving the command.

✓ Request Payload

Parameter	Format	Description
Confirmation Code	int	Confirmation code managed by the initiator

✓ Response Payload

None

Frame Example:：

T> FE000D010C01005E5253541D34
R< FE0009018C01004F1C

4.1.3. Buzzer Beep（0x10）

This command is used to sound the buzzer once.

✓ Request Payload

None

Parameter	Format	Description
Beep Duration	int	Controls the buzzer sounding time, Unit: ms

✓ Response Payload

None

Frame Example：

T> FE000D010F1000000000322AD1
R< FE0009018F1000260E

4.1.4. Configure GPIO Mode （0x11）

This command is used to configure the GPIO input/output mode.。

Only supported by some devices; most devices have fixed GPIO modes that cannot be modified.

✓ Request Payload

Parameter	Format	Description
GPIO Mask	int	Each GPIO occupies 1 BIT, bit0 is GPIO1, bit1 is GPIO2...
GPIO Mode	int	Each GPIO occupies 1 BIT, 1 for Input mode, 0 for Output mode; bit0 is GPIO1 mode, bit1 is GPIO2 mode...

✓ Response Payload

None

Frame Example：

T> FE0011010E1100000000001000000015013
R< FE0009018E1100220F

4.1.5. Query GPIO Mode (0x12)

This command is used to query the GPIO input/output mode.

✓ Request Payload

None

✓ Response Payload

Parameter	Format	Description
GPIO Mask	int	Each GPIO occupies 1 BIT, bit0 is GPIO1, bit1 is GPIO2...
GPIO Mode	int	Each GPIO occupies 1 BIT, 1 for Input mode, 0 for Output mode; bit0 is GPIO1 mode, bit1 is GPIO2 mode...

Frame Example:

T> FE0009010C12002266

R< FE0011018C1200000000FF0000000FD9F5

4.1.6. Configure GPIO Level (0x13)

This command is used to configure the output level state for GPIOs in output mode.

Configuring the level for GPIOs in input mode will have no effect.

✓ Request Payload

Parameter	Format	Description
GPIO Mask	int	Each GPIO occupies 1 BIT, bit0 is GPIO1, bit1 is GPIO2...
GPIO Level	int	Each GPIO occupies 1 BIT, 1 for High level, 0 for Low level; bit0 is GPIO1 output level, bit1 is GPIO2 output level...
Duration	int	0 means the GPIO output level is maintained indefinitely; Non-zero means reverting to the previous output state after the duration (ms).

✓ Response Payload

Parameter	Format	Description
GPIO Mask	int	Each GPIO occupies 1 BIT, bit0 is GPIO1, bit1 is GPIO2...
GPIO Actual Level	int	Each GPIO occupies 1 BIT, 1 for High level, 0 for Low level; bit0 is GPIO1 actual level, bit1 is GPIO2 actual level...

Frame Example:

T> FE0015011113000000001000000001000000000DDCC
R< FE0011019113000000001000000000006717

4.1.7. Query GPIO Level (0x14)

This command is used to query the GPIO level status.

✓ Request Payload

None

✓ Response Payload

Parameter	Format	Description
GPIO Mask	int	Each GPIO occupies 1 BIT, bit0 is GPIO1, bit1 is GPIO2...
GPIO Level	int	Each GPIO occupies 1 BIT, 1 for High level, 0 for Low level; bit0 is GPIO1 input level, bit1 is GPIO2 input level...

Frame Example:

T> FE000901131400E792
R< FE0011019314000000000FF0000000085AB

4.1.8. Get WiFi List (0x50)

This command is used to initiate a WiFi scan and obtain the WiFi results。

✓ Request Payload

Parameter	Format	Description
index	int	Request sequence number. When the Host

		sends sequence number 0, the Reader starts WiFi scan and returns information. Subsequently, the Host requests remaining AP information by sequence number.
count	short	Maximum number of AP information to send in a single response.

✓ Response Payload

Parameter	Format	Description
index	int	If AP information is uploaded, the returned sequence number is the same as the request sequence number; if no AP information is uploaded, ... (Note: Original text incomplete, implies different behavior)
count	short	Number of AP information sent this time; returning 0 indicates transmission is complete.
The following parameters are present only when AP information is uploaded		
id	short	AP ID
SSID	string	AP SSID
BSSID	byte[6]	AP BSSID
Channel	short	Channel used by the AP
RSSI	short	AP signal strength, Unit: 0.01dBm
Security	short	Encryption method used by the AP

Frame Example:

T> FE000F01185000000000000013DA3

R< FE00090198500E0CFF

4.1.9. Control Peripheral (0x54)

This command is used to control device peripherals, such as PSAM, universal serial port, 485, etc.

✓ Request Payload

Parameter	Format	Description
Peripheral Type	byte	0: Serial Port 1: Wiegand 2: PSAM 3: ADC 4: VOUT
Peripheral ID	byte	Peripheral ID, numbered separately for each peripheral type.
Control Code	int	0: Send Data Others: Control code, varies by device type.
Data	array	Data to be sent to the peripheral.

✓ Response Payload

Parameter	Format	Description
Peripheral Type	byte	0: Serial Port 1: Wiegand 2: PSAM
Peripheral ID	byte	Peripheral ID, numbered separately for each peripheral type.
Control Code	int	0: Send Data Others: Control code, varies by device type.
Return Data	array	Data returned by the peripheral.

Frame Example:
Omitted

4.1.10. Firmware Upgrade (0x5F)

This command is used to upgrade the firmware.

✓ Request Payload

Parameter	Format	Description
Index	int	Request sequence number. When the Host sends sequence number 0, the Reader enters upgrade mode, then the Host sends firmware data sequentially by

		sequence number. When the Host sends sequence number 0x7FFFFFFF, the Reader performs firmware verification and exits upgrade mode.
Firmware	array	Firmware data.

✓ Response Payload

Parameter	Format	Description
Index	int	Response sequence number. Must be the same as the request sequence number.

Frame Example:
Omitted

4.2. Parameter Management Commands[0x60~0x6F]

4.2.1. Write Parameters (0x60)

This command is used to modify parameters.

✓ Request Payload

Parameter	Format	Description
type	byte	Parameter Type: 0x00: Device Parameters 0x01: RFID Parameters 0x02: UHF Parameters 0x03: HF Parameters
id	byte	Parameter ID
save	bool	Whether to save to Flash
param	array	Parameter value. Definitions for each parameter type are detailed in the appendix.

✓ Response Payload

Parameter	Format	Description
type	byte	Parameter Type, same as the request frame
id	byte	Parameter ID, same as the request frame

Frame Example:
T> FE000E01346000002F0101016582
R< FE000B01B46000002F5A84

4.2.2. Read Parameters（0x61）

This command is used to query parameters.。

✓ Request Payload

Parameter	Format	Description
type	byte	Parameter Type: 0x00: Device Parameters 0x01: RFID Parameters 0x02: UHF Parameters 0x03: HF Parameters
id	byte	Parameter ID

✓ Response Payload

Parameter	Format	Description
type	byte	Parameter Type, same as the request frame
id	byte	Parameter ID, same as the request frame
param	array	Parameter value. Definitions for each parameter type are detailed in the appendix.

Frame Example:
T> FE000B01366100002F4A63
R< FE000D01B66100002F01016CA3

4.2.3. Reset Parameters（0x62）

This command is used to reset the specified/all parameters of the Reader to factory defaults.。

✓ Request Payload

Parameter	Format	Description
Confirmation Code	int	Fixed value: 0x5E524553 ('^RES' in ASCII)
mask	int	Parameter type mask to reset: bit0: Device Parameters bit1: RFID Parameters bit2: UHF Parameters bit3: HF Parameters

✓ Response Payload

None

Frame Example:

T> FE0011013762005E524553000000077CBA

R< FE000901B762002431

4.3. RFID Management Commands [0x70-0x7F]

4.3.1. Stop（0x7F）

This command is used to forcibly stop RF work. Commonly used to stop inventory, tag reading, carrier detection, etc.

✓ Request Payload

None

✓ Response Payload

Frame Example:

T> FE000901387F004675

R< FE000901B87F007D2F

4.4. RFID Tag General Commands [0x80-0x8F]

4.4.1. Inventory Tags（0x81）

This command is used to inventory tags. Tag data is transmitted via active reporting.

✓ Request Payload

Parameter	Format	Description
Option	int	Inventory options, each bit corresponds to a function: bit0: Cache tags (do not transmit tag data) bit1: Request TID transmission bit2: Request frequency transmission bit3: Request phase transmission bit4: Request read count transmission bit5: Request timestamp transmission bit6: Request RSSI transmission bit8: Enable inventory duration option bit9: Enable tag selection option bit10: Enable password option bit11: Enable preset password option bit16: Enable temperature tag option bit17: Enable humidity tag option bit18: Enable luminous tag option bit19: Enable ImpinJ tag option bit20: Enable NXP tag option link. bit24: Enable read tag bit29: Discard tags with RSSI less than specified value bit30: Merge reporting
The following parameters are present only when the corresponding inventory option is enabled		
timeout	int	Read duration, unit: ms

select	int	Tag selection enable, each bit enables a select rule; Select rules are configured by UHF/HF parameters.
passwd	array	Password option: UHF is the access password.
passwd_index	int	Preset password option: Preset password index.
temperature	int[2]	Temperature tag option: int[0]: Temperature tag type int[1]: RFU
humility	int[2]	Humidity tag option: int[0]: Humidity tag type int[1]: RFU
led	int[2]	Luminous tag option: int[0]: Luminous tag type int[1]: Luminous duration, unit: ms
impinj	int	ImpinJ tag option: bit0: Enable TagFocus bit1: Enable FastID bit8: Enable Gen2X
nxp	int	NXP tag option
read	short[3]	Parameters for enabling tag read: Note: Cached read does not support this function. short[0]: Memory bank, see read tag definition
		short[0]: Memory bank, see read tag definition
		short[1]: Word address
		short[2]: Word length
rssi	short	Discard tags with RSSI less than the specified value, unit: 0.01dBm
merge	int	Merge cycle (unit: ms)

✓ Response Payload

None

Frame Example:

T> FE000D013A810000000050CD1B
R< FE000901BA81002381

4.4.2. Read Inventory Cache Data（0x82）

This command is used for the device to return the read tag data.

✓ Request Payload

Parameter	Format	Description
index	int	Tag index

✓ Response Payload

Parameter	Format	Description
index	int	Tag index
Upload Data Mask	int	Upload Data Mask bit0: ant bit1: pc bit2: epc bit3: tid bit4: count bit5: rssi bit6: frequency bit7: phase bit8: timestamp bit9: temperature bit10: humidity
The following parameters are valid only when the corresponding mask bit is set		
ant	byte	Antenna 1: Antenna 1 2: Antenna 2 ...
PC	short	Tag PC
epc	array	EPC data
tid	array	TID data
count	int	Tag read count

rssi	short	RSSI (0.01dBm)
frequency	int	Read frequency (KHz)
phase	short	Read phase (0.1°)
timestamp	long	Timestamp
temperature	short	Temperature (0.01°C)
humidity	short 【原文为 int16， 统一为 short】	Humidity (0.01%)

✓ Response Payload

None

Frame Example:

Omitted

4.4.3. Read Tag（0x83）

This command is used to read a specified memory bank of a tag. Tag data is transmitted via active reporting.。

✓ Request Payload

Parameter	Format	Description
Option	int	Read options, each bit corresponds to a function: bit8: Enable read duration option bit9: Enable tag selection option bit10: Enable password option bit11: Enable preset password option
bank	short	ISO18000-6C / EPC C1G2 0: Reserved 1: EPC 2: TID 3: User Mifare 0x0000~0x00FF: User block 0 ~ 255 0x0100~0x01FF: Value block 0~255

		Others reserved ISO15693 0x0000~0x00FF: Block 0～255 0x0100: DSFID 0x0101: AFI 0x0102: DSAFI?【原文为卡片详细信息，可能为 DSFID；DSAFI 存疑】
wpoint	short	Word address
wlength	short	Word length
The following parameters are present only when the corresponding inventory option is enabled		
timeout	int	Read duration, unit: ms
select	int	Tag selection enable, each bit enables a select rule; Select rules are configured by UHF/HF parameters.
passwd	array	Password option: UHF is the access password.
passwd_index	int	Preset password option: Preset password index.

✓ Response Payload

Parameter	Format	Description
data	array	Read data

Frame Example:
T> FE0017016583000000020000000001000300000004 97CB
R< FE001201E583000854C8F7780A3CADAF70C5

4.4.4. Write Tag (0x84)

This command is used to write to a specified memory bank of a tag.

✓ Request Payload

Parameter	Format	Description
Option	int	Write options, each bit corresponds to a function:

		bit8: Enable write duration option bit9: Enable tag selection option bit10: Enable password option bit11: Enable preset password option
blocksize	short	Block size 0: Disable block write Non-zero: Perform block write with the specified block size (unit: words)
bank	short	ISO18000-6C / EPC C1G2 0: Reserved 1: EPC 2: TID 3: User 0x0100: Write EPC (auto PC) Mifare 0x0000~0x00FF: User block 0 ~ 255 0x0100~0x01FF: Value block 0~255 Others reserved ISO15693 0x0000~0x00FF: Block 0 ~ 255 0x0100: DSFID 0x0101: AFI 0x0102: DSAFI?【原文为卡片详细信息，可能为 DSFID；DSAFI 存疑】
wpoint	short	Word address
wlength	short	Word length
data	array	Content to write, insufficient words are padded with 0 at the end.
The following parameters are present only when the corresponding inventory option is enabled		
timeout	int	Write duration, unit: ms
select	int	Tag selection enable, each bit enables a select rule; Select rules are configured by UHF/HF parameters.
passwd	array	Password option: UHF is the access password.

passwd_index	int	Preset password option: Preset password index.

✓ Response Payload

None

Frame Example:

T>
FE0022016784000000020000000001000000030000000408112233445566778 8B4B3

R< FE000901E78400C0EB

4.4.5. Lock Tag (0x85)

This command is used to lock a specified memory bank of a tag.

✓ Request Payload

Parameter	Format	Description
Option	int	Lock options, each bit corresponds to a function: bit8: Enable lock operation duration option bit9: Enable tag selection option bit10: Enable password option bit11: Enable preset password option
bank	short	ISO18000-6C / EPC C1G2 0: User 1: TID 2: EPC 3: Access Password 4: Kill Password Mifare 0x0000~0x00FF: User block 0 ~ 255 0x0100~0x01FF: Value block 0~255 Others reserved ISO15693

		0x0000~0x00FF: Block 0 ~ 255 0x0101: AFI 0x0102: DSAFI? 【原文为 DSAFI，存疑】
type	short	0: Writable 1: Permanently Writable 2: Password Writable 3: Permanently Unwritable
The following parameters are present only when the corresponding inventory option is enabled		
timeout	int	Lock operation duration, unit: ms
select	int	Tag selection enable, each bit enables a select rule; Select rules are configured by UHF/HF parameters.
passwd	array	Password option: UHF is the access password.
passwd_index	int	Preset password option: Preset password index.

✓ Response Payload

None

Note:

1. TID bank is permanently unwritable.
2. When the Reserved bank (Access Password/Kill Password) is permanently unwritable, it is also unreadable; when it is password writable, it also requires a password to read.

Frame Example:

T> FE001A01698500000006000000000104000100110000000021FB9
R< FE000901E98500E8DB

4.4.6. Kill Tag (0x86)

This command is used to kill a tag.

✓ Request Payload

Parameter	Format	Description
Option	int	Kill options, each bit corresponds to a function: bit8: Enable kill timeout option bit9: Enable tag selection option bit10: Enable password option bit11: Enable preset password option
The following parameters are present only when the corresponding inventory option is enabled		
timeout	int	Kill duration, unit: ms
select	int	Tag selection enable, each bit enables a select rule; Select rules are configured by UHF/HF parameters.
passwd	array	Password option: UHF is the access password.
passwd_index	int	Preset password option: Preset password index.

✓ Response Payload

None

Frame Example:
Omitted

4.5. UHF Special Commands [0x90-0x9F]

4.5.1. VSWR Detection (0x90)

This command is used to detect the antenna Voltage Standing Wave Ratio (VSWR).

✓ Request Payload

Parameter	Format	Description
Antenna ID	byte	Antenna that needs to transmit carrier wave. 1: Antenna 1 2: Antenna 2 ...
Region	byte	Working region, see Appendix 3 for details.
Frequency	int	Unit: kHz
Tx Power	short	Power to transmit, unit: 0.01dBm. Note: Some devices do not support this; they will only use default power for VSWR detection.

✓ Response Payload

Parameter	Format	Description
Frequency	int	Unit: kHz
RL	short	Unit: 0.01dBm

Frame Example:

T> FE0011017990000100000E1AD70CE41089

R< FE000F01F99000000E1AD701F44A6D

4.6. Active Messages [0xF0~0xFF]

Active messages do not require a response.

4.6.1. GPIO Status Change (0xF0)

This command is used by the device to return GPIO change information.

✓ Payload

Parameter	Format	Description
gpi	byte	GPI number 1. GPI1 2. GPI2 3. ...
value	bool	High/Low level

Frame Example:
R< FE000B0100F0000100DD9F

4.6.2. Peripheral Active Data（0xF4）

This command is used by the device to actively send peripheral data.

✓ Payload

Parameter	Format	Description
Peripheral Type	byte	0: Serial Port 1: Wiegand 2: PSAM 3: ADC 4: VOUT
Peripheral ID	byte	Peripheral number, numbered separately for each peripheral type.
Return Data	array	Data returned by the peripheral.

Frame Example:
Omitted

4.6.3. Inventory Statistics Information（0xF8）

This command is used to send the statistical information of the equipment inventory after it is completed.

✓ Payload

Parameter	Format	Description
time	int	Inventory duration, unit: ms

read count	int	Read count
tag count	int	Number of tags

Frame Example:
Omitted

4.6.4. Tag Data (0xF9)

This command is used by the device to return the read tag data.

✓ Payload

Parameter	Format	Description
Upload Data Mask	int	Upload Data Mask bit0: ant bit1: pc bit2: epc bit3: tid bit4: count bit5: rssi bit6: frequency bit7: phase bit8: timestamp bit9: temperature bit10: humidity bit11: data
Protocol Type	byte	Refer to protocol encoding.
The following parameters are valid only when the corresponding mask bit is set		
ant	byte	Antenna 1: Antenna 1 2: Antenna 2 ...
PC	short	Tag PC
epc	array	EPC data
tid	array	TID data
count	int	Tag read count
rssi	short	RSSI (0.01dBm)
frequency	int	Read frequency (KHz)

phase	short	Read phase (0.1°)
timestamp	long	Timestamp
temperature	short	Temperature (0.01°C)
humidity	short	Humidity (0.01%)
data	array	Data read during tag read operation.

Frame Example:

R<
FE002C0100F900000001370101340000C0000110520008914A0920B010000
0001F4AC386DEFA32A14B48097E6

4.6.5. RFID Stop (0xFF)

This command is used by the device to return the reason for RFID stop.

✓ Payload

Parameter	Format	Description
Stop Reason	byte	0: Active Stop 1: Command Stop 2: Abnormal Stop

Frame Example:

R< FE000A0100FF000111B3

Appendix 1 Parameter Tables

1.1 Device Class Parameters

Param ID	Parameter Name	Value Description
0x01	Mainboard Version Info	byte[4]: Version number
		string: Version description
0x02	Mainboard Serial Number	string: Serial Number
0x03	Ethernet MAC	byte[6]: MAC Address
0x04	WIFI MAC	byte[6]: MAC Address

Param ID	Parameter Name	Value Description
0x05	BT_MAC	byte[6]: MAC Address
0x10	Device Name	string: Device description
0x11	Asset Number	string: Asset ID
0x12	Device Date	int: UTC seconds
		int: UTC nanoseconds
0x13	Device Boot Info	int: Device power-on time (seconds)
		int: Device boot reason
0x14	Boot Event	bool: Boot event enable
		array: Command (complete command, supports multiple)
0x1E	Device Temperature	int: Device temperature (0.01°C)
0x1F	Device SOC	int: Device SOC (0.01%)
0x20	Device Address	byte: Address, 1~250
0x21	RS232 Attributes	int: Baud rate
		byte: Data bit length, can only be 8
		byte: Stop bit length, can be 0,1,2
		byte: Parity bit, 0 N, 1 O, 2 E
		byte: Reserved, fill with 0
0x22	RS485 Attributes	int: Baud rate
		byte: Data bit length, can only be 8
		byte: Stop bit length, can be 0,1,2
		byte: Parity bit, 0 N, 1 O, 2 E
		byte: Reserved, fill with 0
0x24	Ethernet Network Attributes	bool: DHCP enable
		byte[4]: IP Address
		byte[4]: Subnet Mask
		byte[4]: Gateway
		byte[4]: DNS
		byte: Working Mode, 0 TCP Server 1 TCP Client 2 UDP Server 3 UDP Client
		short: Local Port
		string: Remote Address
		short: Remote Port
0x25	WiFi Network Attributes	bool: DHCP enable
		byte[4]: IP Address
		byte[4]: Subnet Mask
		byte[4]: Gateway

Param ID	Parameter Name	Value Description
		byte[4]: DNS
		byte: Working Mode, 0 TCP Server 1 TCP Client 2 UDP Server 3 UDP Client
		short: Local port
		string: Remote address
		short：Remote port
0x26	WiFi Connect AP	string: SSID
		string: Password
0x27	WiFi Connection Info	string: SSID
		bool: Connection Status
0x28	Bluetooth Attributes	
0x2F	Default Connection	byte: Default connection 0：RS232 1：RS485 2：RS232 3：Ethernet 4: WiFi 5: BT
0x51	Buzzer Parameters	bool: Buzzer enable
		short: Buzzer active duration
		short: Buzzer idle duration
0x52	Comm LED Parameters	bool: Communication LED enable
		short: Communication LED active duration
		short: Communication LED idle duration
0x53	Tag LED Parameters	bool: Tag LED enable
		short: Tag LED active duration
		short: Tag LED idle duration
0x60	GPIO1 Interrupt Event	byte: Edge type 0: No trigger 1: Rising edge trigger 2: Falling edge trigger

Param ID	Parameter Name	Value Description
		3: Either edge trigger
		array: Command (complete command, supports multiple)
0x61	GPIO2 Interrupt Event	byte: Edge type 0: No trigger 1: Rising edge trigger 2: Falling edge trigger 3: Either edge trigger
		array: Command (complete command, supports multiple)
0x62	GPIO3 Interrupt Event	byte: Edge type 0: No trigger 1: Rising edge trigger 2: Falling edge trigger 3: Either edge trigger
		array: Command (complete command, supports multiple)
0x63	GPIO4 Interrupt Event	byte: Edge type 0: No trigger 1: Rising edge trigger 2: Falling edge trigger 3: Either edge trigger
		array: Command (complete command, supports multiple)
0x64	GPIO5 Interrupt Event	byte: Edge type 0: No trigger 1: Rising edge trigger 2: Falling edge trigger 3: Either edge trigger
		array: Command (complete command, supports multiple)
0x65	GPIO6 Interrupt Event	byte: Edge type

Param ID	Parameter Name	Value Description
		0: No trigger 1: Rising edge trigger 2: Falling edge trigger 3: Either edge trigger
		array: Command (complete command, supports multiple)
0x66	GPIO7 Interrupt Event	byte: Edge type 0: No trigger 1: Rising edge trigger 2: Falling edge trigger 3: Either edge trigger
		array: Command (complete command, supports multiple)
0x67	GPIO8 Interrupt Event	byte: Edge type 0: No trigger 1: Rising edge trigger 2: Falling edge trigger 3: Either edge trigger
		array: Command (complete command, supports multiple)

1.2 RFID Class Parameters

Param ID	Parameter Name	R/W	Value Description
0x01	RFID version information	RO	byte[4]: Version No.
			string: Version Description
0x02	RFID SN	RO	stirng:Series No.
0x0F	RFID Properties	RO	int: Supported protocol mask bit0: ISO18000-6B bit1: ISO18000-6C/EPC C1G2 bit2: ISO18000-6D bit4: GB29768 bit5: GJB7377.1 bit16: ISO14443A bit17: ISO14443B bit18: ISO15693
			byte: Number of antennas
			short: Maxpower(0.01dBm)
			short: Minpower(0.01dBm)
			short: Default power(0.01dBm)
			short: power step (0.01dBm)
0x10	RFID Current Protocol	RW	int: Configured working protocol mask, depending on device type, supports one or multiple simultaneous (inventory) work bit0: ISO18000-6B bit1: ISO18000-6C/EPC C1G2 bit2: ISO18000-6D bit4: GB29768 bit5: GJB7377.1 bit16: ISO14443A bit17: ISO14443B bit18: ISO15693
0x1E	RFID temperature	RO	short: Device temperature (0.01°C)
0x20	RFID Idle Mode	RW	bool: Enable
			short: Cycle (ms)
			byte: Idle ratio 0~100
0x50	RFID EAS Event	RW	bool: Enable

Param ID	Parameter Name	R/W	Value Description
			short: Data source: ISO18K6C: 0: EPC 1: TID
			short: Start bit address, -1 matches end data
			array: Mask
			array: Expected value
			array: Match execution commands (complete commands, multiple supported)
			array: Mismatch execution commands (complete commands, multiple supported)
0x60	Wiegand Output	RW	bool: Enable
			short: Data source: ISO18K6C: 0: EPC 1: TID Others: 0: Tag UID
			short: Start bit address; -1 means send end data
			byte: Data bit length
0x61	Special Data Format Output	RW	bool: Enable
			short: Data source: ISO18K6C: 0: EPC 1: TID Others: 0: Tag UID
			short: Start bit address, -1 send end data
			short: Data bit length
			byte: Output format 0: Raw output 1: Convert to hex string output
			byte: Check method

Param ID	Parameter Name	R/W	Value Description
			int：Additional data mask bit0: ant bit6: rssi bit7: frequency bit8: phase bit29: SN bit30: AssetID
			array: Frame header
			array: Frame tail
			array: Additional data separator
0x62	特殊格式输出 **和上面重复了?**	RW	bool: Enable
			short: Data source: ISO18K6C: 0: EPC 1: TID Others: 0: Tag UID
			byte: Output format 0: Raw output 1: Convert to hex string output
			Frame tail
0x80	RFID Antenna status	RO	int: Antenna connection mask, bit0 Antenna 1 status, bit1 Antenna 2 status
0x81	Antenna Config	RW	int Working antenna mask, bit0 Antenna 1, bit1 Antenna 2
			short: Maximum antenna working duration (ms)
0x82	All Antenna power setting	WO	short: Read power(0.01dBm)
			short: Write power(0.01dBm)
0x90	Antenna1power	RW	short: Read power(0.01dBm)
			short: Write power(0.01dBm)
0x91	Antenna2power	RW	short: Read power(0.01dBm)
			short: Write power(0.01dBm)
0x92	Antenna3power	RW	short: Read power(0.01dBm)
			short: Write power(0.01dBm)
0x93	Antenna4power	RW	short: Read power(0.01dBm)
			short: Write power(0.01dBm)
0x94	Antenna5power	RW	short: Read power(0.01dBm)
			short: Write power(0.01dBm)

Param ID	Parameter Name	R/W	Value Description
0x95	Antenna6power	RW	short: Read power(0.01dBm)
			short: Write power(0.01dBm)
0x96	Antenna7power	RW	short: Read power(0.01dBm)
			short: Write power(0.01dBm)
0x97	Antenna8power	RW	short: Read power(0.01dBm)
			short: Write power(0.01dBm)
0x98	Antenna9power	RW	short: Read power(0.01dBm)
			short: Write power(0.01dBm)
0x99	Antenna10power	RW	short: Read power(0.01dBm)
			short: Write power(0.01dBm)
0x9A	Antenna11power	RW	short: Read power(0.01dBm)
			short: Write power(0.01dBm)
0x9B	Antenna12power	RW	short: Read power(0.01dBm)
			short: Write power(0.01dBm)
0x9C	Antenna13power	RW	short: Read power(0.01dBm)
			short: Write power(0.01dBm)
0x9D	Antenna14power	RW	short: Read power(0.01dBm)
			short: Write power(0.01dBm)
0x9E	Antenna15power	RW	short: Read power(0.01dBm)
			short: Write power(0.01dBm)
0x9F	Antenna16power	RW	short: Read power(0.01dBm)
			short: Write power(0.01dBm)
0xF0	Preset Key 1	WO	Preset Key，UHF 4 bytes,
0xF1	Preset Key 2	WO	Preset Key
0xF2	Preset Key 3	WO	Preset Key
0xF3	Preset Key 4	WO	Preset Key
0xF4	Preset Key 5	WO	Preset Key
0xF5	Preset Key 6	WO	Preset Key
0xF6	Preset Key 7	WO	Preset Key
0xF7	Preset Key 8	WO	Preset Key
0xF8	Preset Key 9	WO	Preset Key
0xF9	Preset Key 10	WO	Preset Key
0xFA	Preset Key 11	WO	Preset Key
0xFB	Preset Key 12	WO	Preset Key
0xFC	Preset Key 13	WO	Preset Key
0xFD	Preset Key 14	WO	Preset Key
0xFE	Preset Key 15	WO	Preset Key
0xFF	Preset Key 16	WO	Preset Key

1.3 UHF Class Parameters

Param ID	Parameter Name	R/W	Value Description
0x0F	UHF Properties	RO	bitmap[256]: Supported working regions bitmap
			bitmap[256]: Supported working regions bitmap
0x10	UHF Frequency Parameters	RW	byte: region
			byte: Auto frequency hopping mode 0: Manual frequency hopping 1: Auto frequency hopping 2: Random frequency hopping
			array: Manual frequency hopping table
			short: Frequency hopping interval (ms)
			int: Start frequency (kHz)
			short: Frequency step (kHz)
			byte: Number of channels
			int: Start frequency 2 (kHz)
			short: Frequency step 2 (kHz)
			byte: Number of channels 2
0x60	UHF EPC Baseband Parameters	RW	byte: Working mode
			byte: session
			byte: Target
			byte: Initial Q value
			byte: Minimum Q value
			byte: Maximum Q value
0x61	UHF EPC lgorithm Parameters	RW	byte: Algorithm type
			byte: Algorithm parameter 1
			short: Algorithm parameter 2
			int: Algorithm parameter 3
0x80	Tag Select Rule 1	RW	byte: AND (0) OR (1) relation
			short: Memory bank to match ISO18K6C 0: Reserved 1: EPC 2: TID 3: User
			short: Start bit address to match
			short: Bit length to match
			array: Data to match
0x81	Tag Select Rule 2	RW	(Structure same as 0x80)

Param ID	Parameter Name	R/W	Value Description
0x82	Tag Select Rule 3	RW	(Structure same as 0x80)
0x83	Tag Select Rule 4	RW	(Structure same as 0x80)
0x84	Tag Select Rule 5	RW	(Structure same as 0x80)
0x85	Tag Select Rule 6	RW	(Structure same as 0x80)
0x86	Tag Select Rule 7	RW	(Structure same as 0x80)
0x87	Tag Select Rule 8	RW	(Structure same as 0x80)

Appendix 2 Working Regions Table

The list of working regions supported by the device is determined by the factory settings.

ID	Description	Channel Count
0	CMIIT 920~925MHz	16
1	FCC 902~928MHz	50

ID	Description	Channel Count
2	ETSI 866~867MHz	4
3	ETSI_HIGH 916~918MHz	3
4	ETSI_HYBRID 866~867MHz 916~918MHz	7
5	KOR 917~921MHz	6
6	JPN_NO_LBT 916~921MHz	4
7	JPN_LBT 916~921MHz	6
8	JPN_NO_LBT_19 916~924MHz	19
9	TWN 920~927MHz	14
10	MYS 919~923MHz	8
11	RUS_LOW 866~867MHz	5
12	RUS_HIGH 916~920MHz	4
13	AUS 920~926MHz	10
14	IND 865~867MHz	4
15	BRA 902~907.5MHz 915~928MHz	35
16	VIE 918~923MHz	6
17	ISR 915~917MHz	1
18	ZAF 916~920MHz	17
19	SGP 920~925MHz	10
20	THA 920~925MHz	10

ID	Description	Channel Count
240	TEST1 800~1000MHz	197
241	TEST2 840~960MHz	117
242	TEST3 860~960MHz	11
243	TEST4 860~930MHz	8

Appendix 3 Device Status Codes

Device Status Codes

Definition	Description
SUCCESS	Success
COMMAND_INVALID_TYPE	Invalid command type
COMMAND_INVALID_LENGTH	Invalid command length
COMMAND_INVALID_PARAM	Invalid command parameter
COMMAND_EXEC_ERROR	Command execution error
COMMAND_NO_RESPONSE	Command no response
GPIO_OVER_RANGE	GPIO index out of range
GPIO_INVALID_MODE	Invalid GPIO mode
GPIO_OTHER_ERROR	Other GPIO error
PARAM_INVALID_TYPE	Invalid parameter type
PARAM_INVALID_VALUE	Invalid parameter value
PARAM_INVALID_LENGTH	Invalid parameter length
PARAM_GET_FAILED	Get parameter failed
PARAM_SET_FAILED	Set parameter failed
PARAM_SAVE_FAILED	Save parameter failed
PARAM_OTHER_ERROR	Other parameter error
TAG_OTHER_ERROR	Tag returned other error
TAG_NOT_SUPPORT	Tag returned not supported
TAG_INSUFFICIENT_PRIVILEGES	Tag returned insufficient privileges
TAG_MEMORY_OVERRUN	Tag returned memory overrun
TAG_MEMORY_LOCKED	Tag returned memory locked
TAG_CRYPTOGRAPHIC_SUITE_ERROR	Tag returned cryptographic suite error
TAG_COMMAND_NOT_ENCAPSULATED	Tag returned command not

Definition	Description
	encapsulated
TAG_RESPONSEBUFFER_OVERFLOW	Tag returned response buffer overflow
TAG_SECURITY_TIMEOUT	Tag returned security timeout
TAG_INSUFFICIENT_POWER	Tag returned insufficient power
TAG_LOST	Tag lost
TAG_NON_SPECIFIC_ERROR	Non-specific error
RFID_INVALID_REGION	Invalid RFID region
RFID_INVALID_VALUE	Invalid RFID value
RFID_EXEC_ERROR	RFID execution error
RFID_NO_ACK	RFID no acknowledgment
RFID_INTERNEL_ERROR	RFID internal error
RFID_OTHER_ERROR	Other RFID error
DEVICE_BUSY	Device busy
INTERNAL_ERROR	Internal error

Appendix 4 CRC Calculation

```
uint16_t crc16_ccitt_false(const uint8_t *data, uint16_t len) {
    uint16_t crc = 0xFFFF;
    uint16_t i, j;

    for (i = 0; i < len; i++) {
        crc ^= (uint16_t)data[i] << 8;
        for (j = 0; j < 8; j++) {
            if (crc & 0x8000) {
                crc = (crc << 1) ^ 0x1021;
            } else {
                crc <<= 1;
            }
        }
    }

    return crc;
}
```

```
/**
 * @brief Calculate CRC-16/CCITT checksum for data
 * @param data Pointer to the data for CRC calculation
 * @param len Data length
 * @return Calculated CRC-16/CCITT checksum
 */
uint16_t crc16_ccitt_false(const uint8_t *data, uint16_t len) {
	uint16_t crc = 0xFFFF;
	uint16_t i, j;
	for (i = 0; i < len; i++) {
		crc ^= (uint16_t)data[i] << 8;
		for (j = 0; j < 8; j++) {
			if (crc & 0x8000) {
				crc = (crc << 1) ^ 0x1021;
			} else {
				crc <<= 1;
			}
		}
	}
	return crc;
}
```